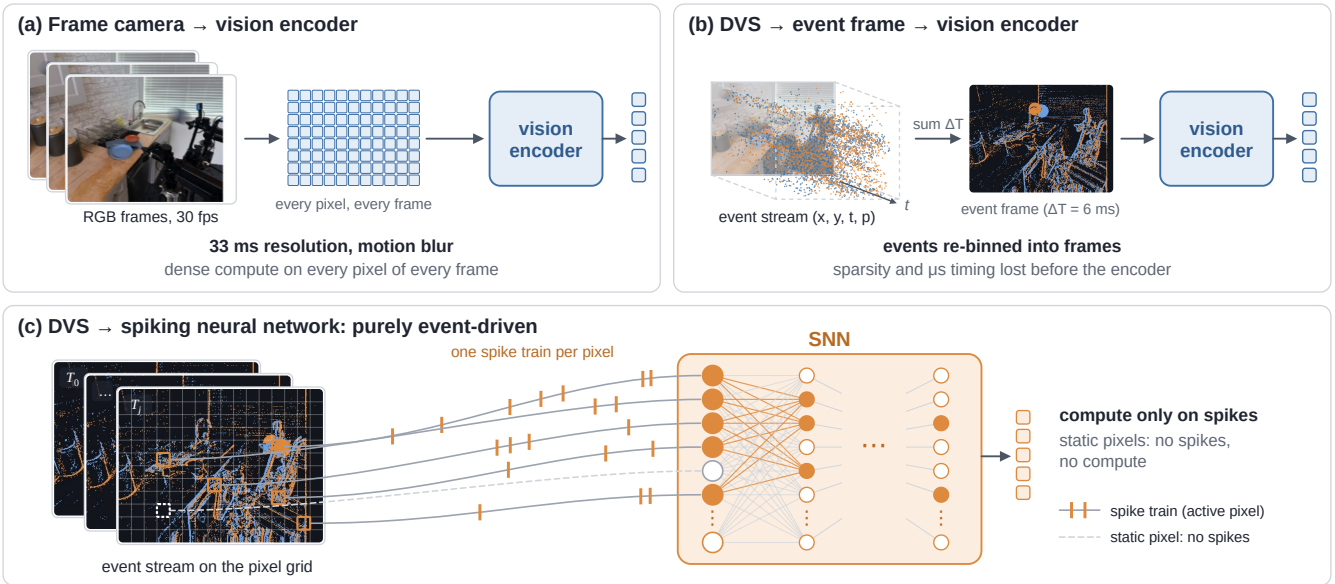

(a) Frame camera → vision encoder
RGB frames, 30 fps
every pixel, every frame
vision encoder
33 ms resolution, motion blur
dense compute on every pixel of every frame
(b) DVS → event frame → vision encoder
sum ΔT
t
event stream (x, y, t, p)
event frame (ΔT = 6 ms)
vision encoder
events re-binned into frames
sparsity and µs timing lost before the encoder
(c) DVS → spiking neural network: purely event-driven
one spike train per pixel
SNN
event stream on the pixel grid
compute only on spikes
static pixels: no spikes, no compute
spike train (active pixel)
static pixel: no spikes